

Roll No.

TOE-801

B. TECH. (EE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
EXPERT SYSTEM AND FUZZY LOGIC

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) What is an Expert Systems ? What are the characteristics/attributes of Expert Systems ? (CO1)
- (b) Why should we use Expert Systems ? What are the limitations of expert systems ? (CO1)
- (c) What is a expert system shell ? Explain the following : (CO1, CO5)
 - (i) MYCIN
 - (ii) EMYCIN
 - (iii) ROSIE

P. T. O.

2. (a) Draw the general structure of an expert system. Also explain its components. (CO1, CO3)
- (b) What is a rule-based expert system ? Give *one* example of rule-based reasoning in expert system. (CO1, CO3)
- (c) What are the differences between a decision support system and an expert system ? (CO1, CO3)
3. (a) What is knowledge elicitation ? Explain the different knowledge elicitation techniques in expert system. (CO1)
- (b) Explain the techniques for representing the Expert Knowledge in expert system. (CO1)
- (c) Explain the use of fuzzy set theory in expert systems by sketching the example of "Linguistic Description of Human Judgments". (CO5, CO3, CO4)
4. (a) For the following navigation problem design a Mamdani fuzzy logic controller. Let us consider a problem scenario related to navigation of a mobile robot in presence of four moving obstacle. Obstacle 2 is found to be the most critical one. The aim is to develop a fuzzy logic-based motion planner that will be able to generate the collision free path for the robot. There are two inputs, namely the distance between the robot and the obstacle (D), and the angle ($\angle$ GSO₂) for the motion planner and it will generate one output, that is, deviation. Distance is represented using four linguistic terms, namely very near (VN), Near (NR), far (FR) and very far (VFR). Whereas the input (i. e., angle) and

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the output (i. e., deviation are expressed with the help of five linguistic terms such as Ahead (A), Ahead Left (AL), Left (LT), Ahead Right (AR) and Right (RT). Determine the output. (CO2, CO3, CO4, CO5)

- (b) For the problem scenario mentioned in question 4 (a), determine the output if distance $D = 1.04$ m and angle ($\angle GSO_2$) = 30, using Mamdani approach. Use any *one* of the defuzzification method. (CO2, CO4)
- (c) What is fuzzy logic and crisp logic ? How do we represent a crisp and fuzzy set ? Also explain the discrete and continuous fuzzy set with examples. (CO2)
5. (a) Explain different defuzzification methods used in fuzzy logic. (CO2)
- (b) What is fuzzy logic ? What is linguistic fuzzy modelling and precise fuzzy modelling ? (CO2)
- (c) Draw the block diagram of fuzzy logic controller. Explain the function of each component. Also give a suitable example. (CO2)

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**B. TECH. (EE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
INTELLIGENT SENSORS AND INSTRUMENTATION**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) What do you understand by a sensor ? Briefly explain the principles of sensors. List the various categories of sensors. (CO1)
- (b) How does transducers differ from actuators ? Justify with suitable explanation. (CO1)
- (c) Compare the conventional sensors and smart sensors. (CO1)
2. (a) Explain briefly about the static and dynamic characteristics of sensors. (CO2)
- (b) What is meant by characterization of a sensor ? Explain briefly about mechanical and thermal characterization of a sensor. (CO2)

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- (c) Why is the resolution of a transducer an important characteristic ?
Explain in detail. (CO2)
3. (a) Classify the various techniques used to fabricate MEMS based sensors.
Explain any *one* technique in detail. (CO3, CO4)
- (b) Illustrate the implementation of sensors in Internet of Things.
(CO3, CO4)
- (c) Explain in your own word the significance and role of computer and
interfacing of sensors with computers in the intelligent instrumentation.
(CO3, CO4)
4. (a) Summarize the key features of IEEE 1451 standard used to design smart
sensors. (CO5)
- (b) What are the various modules of IEEE 1451 standard ? Explain briefly.
(CO5)
- (c) What are the *five* different parts of TEDS ? Explain briefly. (CO5)
5. (a) What is the importance of measurement in industries ? What are the
various ways and means available in industries to measure various
parameters ? (CO6, CO3)
- (b) Describe in detail, how the revolution in the sensor fabrication
technologies helped in industries. (CO6, CO3)
- (c) Give suitable comparison conventional instrumentation and intelligent
instrumentation. (CO6, CO3)

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B. TECH. (EE) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2022
WIND AND SOLAR ENERGY SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write a short note on variation of wind speed with height. With the help of a labelled diagram discuss Ekman layer and planetary boundary layer. (CO1)

(b) Following data were recorded for a two blade HAWT : (CO1)

Average free wind speed at a standard height of 10 m = 8 m/s,

$$\alpha = 0.13$$

$$\text{Air density} = 1.226 \text{ kg/m}^3$$

Hub height from the ground = 80 m

Rotor diameter = 60 m

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Downstream wind velocity is half that of upstream wind.

Find :

- (i) Power available in the wind
 - (ii) Power extracted by the turbine
 - (iii) Axial force on the turbine
- (c) Define and explain the terms : Solidity, Rotor swept area, Tip speed ratio, Power coefficient in wind energy systems. (CO1)
2. (a) Explain the need for the control systems in wind Turbines. What is Pitch angle control. (CO2)
- (b) An aero-generator, installed at seashore generates an output of 1200 W at wind speed of 6 m/s at a temperature of 27°C. What will be the output, if the same aero-generator is installed on the top of a hill where the temperature is 15°C, pressure is 0.85 atmospheric and wind speed is 8 m/s ? (CO2)
- (c) Explain clearly how a DFIG works satisfactorily in a variable speed wind turbine system over a wide speed range. (CO2)
3. (a) Define solar irradiance, solar constant, extraterrestrial and terrestrial radiations. What is the standard value of solar constant ? (CO3, CO4)
- (b) What maximum wavelength can a photon have to create hole-electron pairs in silicon ? What minimum frequency is that ? Silicon has a band gap of 1.12 eV ($1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$). (CO3, CO4)
- (c) Define and explain the terms : (CO3, CO4)
- (i) Cell efficiency and
 - (ii) Fill factor

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4. (a) Draw and explain an equivalent circuit of a practical solar PV cell. (CO5)
- (b) On the basis of important features, differentiate between 'Stand Alone Solar Power System' and 'Grid Connected Solar Power System. (CO5)
- (c) What is the importance of MPPT in a solar PV system ? Explain the various strategies used for operation of an MPPT. (CO5)
5. (a) What are the interfacing requirements and the conditions for synchronizing of such a wind generator with the grid ? (CO5, CO3)
- (b) What are the power quality issues in the interconnected systems of solar PV and wind system ? Elaborate. (CO5, CO3)
- (c) What are the various technical and non-technical challenges and issues associated with the grid integration of various renewable energy sources, particularly solar photovoltaic and wind energy conversion systems. Propose some possible solution of the aforesaid problems. (CO5, CO3)